Observation - Week 04

Topic: MongoDB and Mongoose

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Aim

To connect Node.js to MongoDB and perform CRUD operations on a User collection using a Mongoose model and controller.

Concepts Covered

- NoSQL documents and collections
- Connecting with mongoose.connect()
- Schemas, models and built-in validation (required, min, enum, unique)
- CRUD with create, find, findByIdAndUpdate and findByIdAndDelete

Observations

- A schema adds structure and validation on top of MongoDB's flexible documents.
- runValidators: true is needed for validation to run on update queries.
- unique: true creates an index; a duplicate email is rejected by the database.
- An invalid ObjectId makes findById throw, so the controller returns 400 instead of crashing.

Viva Questions

1 What is the difference between SQL and NoSQL?

2 What is a Mongoose schema?

3 What does { new: true } do?

4 What is an ObjectId?

Reference Files

- src/dbConfig.js
- src/models/User.js
- src/controllers/userController.js
- screenshots/mongodb-compass-db.png
- screenshots/crud-operations-output.png